

CHURCHILL FALLS, NL, Canada

WMO: 710020

Lat: 53.562N

Lon: 64.106W

Elev: 1442

StdP: 13.95

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.3	-26.3	-39.3	0.6	-30.0	-35.5	0.8	-25.9	25.9	0.2	23.3	1.7	4.8	270	0.776

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	77.0	61.0	73.0	59.2	69.5	57.8	63.9	72.3	61.7	69.2	59.8	66.6	9.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	84.4	66.0	58.9	78.3	64.3	56.8	72.5	62.8	30.0	72.1	28.4	69.3	27.0	66.5	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	22.9	20.2	18.0	DB	-37.3	83.6	3.9	2.8	-40.1	85.6	-42.4	87.2	-44.6	88.8	-47.4	90.8
(5)				WB	-37.4	67.6	3.9	2.5	-40.2	69.3	-42.5	70.8	-44.7	72.1	-47.6	73.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	-4.8	-3.1	9.1	24.5	39.4	51.0	58.0	56.4	46.7	35.1	20.2	4.4
	DBStd	24.11	13.65	12.93	13.25	10.41	8.16	7.86	5.87	5.93	6.88	7.17	10.14	13.17
	HDD50	8574	1697	1488	1268	765	345	78	7	11	144	465	894	1412
	HDD65	13441	2162	1908	1733	1215	795	425	229	275	551	927	1344	1877
	CDD50	630	0	0	0	0	15	108	254	208	43	2	0	0
	CDD65	26	0	0	0	0	0	5	12	8	1	0	0	0
	CDH74	263	0	0	0	0	11	66	102	73	11	0	0	0
	CDH80	35	0	0	0	0	1	14	12	6	2	0	0	0
Wind	WSAvg	8.4	8.4	8.6	9.3	9.5	8.3	8.3	7.4	7.2	8.3	8.7	8.3	8.1
Precipitation	PrecAvg	39.3	2.8	2.3	2.7	2.8	2.4	3.9	4.6	3.8	4.5	3.3	3.5	2.8
	PrecMax	48.6	5.1	4.8	6.3	4.7	4.2	7.2	8.1	7.7	6.1	6.0	6.1	6.2
	PrecMin	31.8	1.0	0.6	1.2	1.2	1.1	1.4	1.7	1.1	1.9	1.9	0.9	1.4
	PrecStd	4.7	1.3	1.0	1.3	1.1	1.0	1.8	1.5	1.6	1.1	1.0	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	35.4	42.2	53.2	73.9	81.3	81.3	80.5	74.2	59.9	47.0	36.8
		MCWB	33.1	33.6	34.0	40.4	55.4	61.4	63.0	64.2	61.1	52.4	45.3	35.3
	2%	DB	30.0	31.1	36.9	46.9	66.0	74.6	76.7	75.7	66.6	53.8	40.6	33.3
		MCWB	28.7	29.5	32.7	37.4	50.2	58.0	61.1	62.3	57.0	49.6	38.9	32.4
	5%	DB	24.0	24.5	33.3	42.9	59.4	69.5	73.0	71.0	61.9	49.7	36.3	29.8
		MCWB	22.9	22.6	30.2	35.9	46.8	55.9	59.2	59.9	54.4	45.7	34.6	28.8
	10%	DB	17.1	18.0	29.6	39.5	53.7	65.3	69.4	67.4	58.1	45.9	33.3	24.0
		MCWB	15.9	16.6	26.5	34.2	44.4	54.0	58.3	58.3	52.0	42.7	32.1	22.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.1	34.2	36.8	42.4	56.4	64.7	66.9	67.2	63.2	53.8	45.5	36.0
		MCDB	33.8	35.2	39.9	48.4	70.4	75.5	71.9	75.1	72.1	56.6	46.6	36.9
	2%	WB	28.9	29.4	33.1	39.2	51.7	60.8	63.5	64.1	58.8	50.1	39.1	32.5
		MCDB	30.1	30.9	36.0	44.6	62.2	69.7	71.5	72.2	64.2	53.5	40.1	33.2
	5%	WB	23.0	22.6	30.2	36.8	48.3	58.2	61.8	61.5	55.9	46.4	34.8	28.6
		MCDB	24.0	24.5	33.6	42.0	57.1	66.0	69.2	67.8	61.3	48.6	35.9	29.6
	10%	WB	16.1	16.5	26.8	34.7	45.0	55.6	60.1	59.2	52.7	42.7	32.3	22.9
		MCDB	17.1	18.0	29.6	39.3	53.5	63.0	66.9	65.4	56.9	45.3	33.3	24.0
Mean Daily Temperature Range	5% DB	MDBR	18.1	20.3	21.7	18.6	17.9	18.5	17.2	15.9	14.5	10.6	12.2	15.4
		MCDBR	18.7	19.5	20.8	22.8	29.8	26.0	23.7	22.9	22.1	18.2	13.8	14.9
		MCWBR	18.2	18.3	17.3	16.2	17.5	13.4	11.7	12.3	14.6	14.0	12.9	15.1
	5% WB	MCDBR	18.5	19.2	18.8	18.6	26.1	20.7	18.7	18.7	19.5	15.8	13.3	15.0
		MCWBR	18.1	18.1	16.3	14.0	16.4	12.3	11.2	11.2	14.3	13.2	12.8	15.1
Clear-Sky Solar Irradiance	taub	0.233	0.246	0.268	0.315	0.355	0.360	0.387	0.353	0.331	0.295	0.253	0.232	
	taud	2.233	2.246	2.221	2.163	2.297	2.419	2.366	2.471	2.499	2.532	2.305	2.174	
	Ebn at Noon	241	277	293	290	283	282	272	276	269	258	231	213	
	Edh at Noon	19	26	34	41	38	35	36	30	26	20	18	16	
All-Sky Solar Radiation	RadAvg	321	668	1131	1560	1683	1625	1493	1207	895	499	255	212	
	RadStd	25	56	103	121	145	125	93	78	82	36	33	25	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	+2.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page